

Savitribai Phule Pune University
T.Y.B.Sc. (Computer Science) Sem-VI Practical Examination
CS-369: Practical Course based on CS-365
Object Oriented Programming using Java – II

Practical Duration: 3 Hours Max. Marks: 35

Note: Replace jdbc:mysql://localhost:3306/mydb, root, password with your actual DB credentials. Always add Class.forName("com.mysql.jdbc.Driver"); before getConnection() in JDBC programs.

SLIP 1

Q1. [15 Marks]

Write a Java program to display all the alphabets between 'A' to 'Z' after every 2 seconds.

Solution: AlphaDisplay.java

```
class AlphaDisplay extends Thread {
    public void run() {
        for (char c = 'A'; c <= 'Z'; c++) {
            System.out.println(c);
            try { Thread.sleep(2000); } catch (Exception e) {}
        }
    }
    public static void main(String[] args) {
        new AlphaDisplay().start();
    }
}
```

Q2. [15 Marks]

Write a Java program to accept the details of Employee (Eno, EName, Designation, Salary) from a user and store it into the database. (Use Swing)

Solution: EmployeeForm.java

```
import javax.swing.*; import java.sql.*;
public class EmployeeForm extends JFrame {
    JTextField eno=new JTextField(8), ename=new JTextField(10),
    desg=new JTextField(10), sal=new JTextField(8);
    public EmployeeForm() {
        setLayout(new java.awt.FlowLayout());
        add(new JLabel("Eno:")); add(eno); add(new JLabel("Name:")); add(ename);
        add(new JLabel("Designation:")); add(desg); add(new JLabel("Salary:")); add(sal);
        JButton btn = new JButton("Save");
        btn.addActionListener(e -> {
            try {
                Connection con = DriverManager.getConnection(
                    "jdbc:mysql://localhost:3306/mydb","root","password");
                PreparedStatement ps = con.prepareStatement(
                    "INSERT INTO Employee VALUES(?,?,?,?)");
                ps.setInt(1,Integer.parseInt(eno.getText()));
                ps.setString(2,ename.getText()); ps.setString(3,desg.getText());
                ps.setDouble(4,Double.parseDouble(sal.getText()));
                ps.executeUpdate();
                JOptionPane.showMessageDialog(this,"Saved!"); con.close();
            } catch (Exception ex){ JOptionPane.showMessageDialog(this,ex.getMessage()); }
        });
        add(btn); setSize(320,180); setVisible(true);
        setDefaultCloseOperation(EXIT_ON_CLOSE);
    }
    public static void main(String[] a) throws Exception {
        Class.forName("com.mysql.jdbc.Driver"); new EmployeeForm();
    }
}
```

Q3. Viva [05 Marks]

SLIP 2

Q1. [15 Marks]

Write a Java program to read 'N' names of your friends, store it into HashSet and display them in ascending order.

Solution: HashSetNames.java

```
import java.util.*; import java.util.Scanner;
public class HashSetNames {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter N: "); int n = sc.nextInt();
        Set<String> set = new TreeSet<>(); // TreeSet maintains sorted order
        for (int i = 0; i < n; i++) { System.out.print("Name: "); set.add(sc.next()); }
        System.out.println("Sorted Names: " + set);
    }
}
```

Q2. [15 Marks]

Design a servlet that provides information about a HTTP request from a client, such as IP-Address and browser type. The servlet also provides information about the server on which the servlet is running, such as the operating system type, and the names of currently loaded servlets.

Solution: InfoServlet.java

```
import java.io.*; import javax.servlet.*; import javax.servlet.http.*;
public class InfoServlet extends HttpServlet {
    public void doGet(HttpServletRequest req, HttpServletResponse res)
        throws IOException {
        res.setContentType("text/html");
        PrintWriter pw = res.getWriter();
        pw.println("<h2>Client Info</h2>");
        pw.println("IP Address: " + req.getRemoteAddr() + "<br>");
        pw.println("Browser: " + req.getHeader("User-Agent") + "<br>");
        pw.println("<h2>Server Info</h2>");
        pw.println("OS: " + System.getProperty("os.name") + "<br>");
        pw.println("Server: " + getServletContext().getServerInfo());
    }
}
```

Q3. Viva [05 Marks]

SLIP 3

Q1. [15 Marks]

Write a JSP program to display the details of Patient (PNo, PName, Address, Age, Disease) in tabular form on browser.

Solution: patient.jsp

```
<%@ page import="java.sql.*" %>
<html><body>
<h2>Patient Details</h2>
<table border="1"><tr><th>PNo</th><th>PName</th><th>Address</th><th>Age</th><th>Disease</th></tr>
<%
    Connection con = DriverManager.getConnection(
        "jdbc:mysql://localhost:3306/mydb","root","password");
    ResultSet rs = con.createStatement().executeQuery("SELECT * FROM Patient");
    while(rs.next()) { %>
        <tr><td><%=rs.getInt(1)%></td><td><%=rs.getString(2)%></td>
        <td><%=rs.getString(3)%></td><td><%=rs.getInt(4)%></td>
        <td><%=rs.getString(5)%></td></tr>
    <% } con.close(); %>
</table></body></html>
```

Q2. [15 Marks]

Write a Java program to create LinkedList of String objects and perform the following:

- i. Add element at the end of the list
- ii. Delete first element of the list
- iii. Display the contents of list in reverse order

Solution: LinkedListDemo.java

```
import java.util.*;
public class LinkedListDemo {
    public static void main(String[] args) {
        LinkedList<String> list = new LinkedList<>(
            Arrays.asList("Apple", "Banana", "Cherry"));
        list.addLast("Mango"); // i. Add at end
        list.removeFirst(); // ii. Delete first
        Collections.reverse(list); // iii. Reverse
        System.out.println("Result: " + list);
    }
}
```

Q3. Viva [05 Marks]

Q1. [15 Marks]

Write a Java program using Runnable interface to blink Text on the frame.

Solution: BlinkText.java

```
import javax.swing.*; import java.awt.*;
public class BlinkText extends JFrame implements Runnable {
    JLabel label = new JLabel("BLINKING TEXT", SwingConstants.CENTER);
    public BlinkText() {
        add(label); setSize(300,200); setVisible(true);
        setDefaultCloseOperation(EXIT_ON_CLOSE);
        new Thread(this).start();
    }
    public void run() {
        while(true) {
            label.setVisible(!label.isVisible());
            try { Thread.sleep(500); } catch(Exception e) {}
        }
    }
    public static void main(String[] a) { new BlinkText(); }
}
```

Q2. [15 Marks]

Write a Java program to store city names and their STD codes using an appropriate collection and perform following operations:

- i. Add a new city and its code (No Duplicates)
- ii. Remove a city from the collection
- iii. Search for a city name and display the code

Solution: CitySTD.java

```
import java.util.*; import java.util.Scanner;
public class CitySTD {
    public static void main(String[] args) {
        Map<String,String> map = new HashMap<>();
        map.put("Pune","020"); map.put("Mumbai","022"); map.put("Nagpur","0712");
        Scanner sc = new Scanner(System.in); int ch;
        do {
            System.out.println("1.Add 2.Remove 3.Search 4.Exit");
            ch = sc.nextInt(); sc.nextLine();
            if(ch==1){
                System.out.print("City: "); String c=sc.nextLine();
                System.out.print("Code: "); map.put(c, sc.nextLine());
            } else if(ch==2){
                System.out.print("City to remove: "); map.remove(sc.nextLine());
            } else if(ch==3){
                System.out.print("City to search: ");
                System.out.println(map.getDefault(sc.nextLine(),"Not found"));
            }
        } while(ch!=4);
    }
}
```

Q3. Viva [05 Marks]

Q1. [15 Marks]

Write a Java Program to create the hash table that will maintain the mobile number and student name. Display the details of student using Enumeration interface.

Solution: HashTableDemo.java

```
import java.util.*;
public class HashTableDemo {
    public static void main(String[] args) {
        Hashtable<String,String> ht = new Hashtable<>();
        ht.put("9876543210", "Alice");
        ht.put("9123456789", "Bob");
        ht.put("9000011111", "Charlie");
        Enumeration<String> keys = ht.keys();
        while(keys.hasMoreElements()) {
            String k = keys.nextElement();
            System.out.println("Mobile: "+k+" | Name: "+ht.get(k));
        }
    }
}
```

Q2. [15 Marks]

Create a JSP page for an online multiple choice test. The questions are randomly selected from a database and displayed on the screen. The choices are displayed using radio buttons. When the user clicks on next, the next question is displayed. When the user clicks on submit, display the total score on the screen.

Solution: mcq.jsp

```
<%@ page import="java.sql.*" session="true" %>
<%
String action = request.getParameter("action");
Connection con = DriverManager.getConnection(
    "jdbc:mysql://localhost:3306/mydb","root","password");
if("submit".equals(action)) {
    int score=(Integer)session.getAttribute("score");
    out.println("<h2>Your Score: "+score+"</h2>");
    con.close(); return;
}
Integer qno=(Integer)session.getAttribute("qno");
Integer score=(Integer)session.getAttribute("score");
if(qno==null){ qno=1; score=0; }
String ans=request.getParameter("ans");
String correct=(String)session.getAttribute("correct");
if(ans!=null && correct!=null && ans.equals(correct)) score++;
session.setAttribute("score",score);
ResultSet rs = con.createStatement().executeQuery(
    "SELECT * FROM Questions WHERE qid="+qno);
if(rs.next()) { %>
<form method="post">
<p><%=rs.getString("question")%></p>
<input type="radio" name="ans" value="A"> <%=rs.getString("optA")%><br>
<input type="radio" name="ans" value="B"> <%=rs.getString("optB")%><br>
<input type="radio" name="ans" value="C"> <%=rs.getString("optC")%><br>
<input type="radio" name="ans" value="D"> <%=rs.getString("optD")%><br>
<% session.setAttribute("correct",rs.getString("answer"));
session.setAttribute("qno",qno+1); %>
<input type="submit" name="action" value="next">
<input type="submit" name="action" value="submit">
</form>
<% } con.close(); %>
```

Q3. Viva [05 Marks]

Q1. [15 Marks]

Write a Java program to accept 'n' integers from the user and store them in a collection. Display them in the sorted order. The collection should not accept duplicate elements. (Use a suitable collection). Search for a particular element using predefined search method in the Collection framework.

Solution: SortedSet.java

```
import java.util.*; import java.util.Scanner;
public class SortedSet {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter N: "); int n = sc.nextInt();
        TreeSet<Integer> ts = new TreeSet<>();
        for(int i=0;i<n;i++){ System.out.print("Enter: "); ts.add(sc.nextInt()); }
        System.out.println("Sorted (no duplicates): " + ts);
        System.out.print("Search: "); int key = sc.nextInt();
        System.out.println(ts.contains(key) ? "Found!" : "Not found.");
    }
}
```

Q2. [15 Marks]

Write a Java program to simulate traffic signal using threads.

Solution: TrafficSignal.java

```
public class TrafficSignal extends Thread {
    String[] colors = {"RED","YELLOW","GREEN"};
    int[] times = {3000,1000,3000};
    public void run() {
        while(true) {
            for(int i=0;i<3;i++) {
                System.out.println("Signal: " + colors[i]);
                try { Thread.sleep(times[i]); } catch(Exception e) {}
            }
        }
    }
    public static void main(String[] a) { new TrafficSignal().start(); }
}
```

Q3. Viva [05 Marks]

Q1. [15 Marks]

Write a Java program that implements a multi-thread application that has three threads. First thread generates random integer number after every one second; if the number is even, second thread computes the square of that number and prints it. If the number is odd, the third thread computes the cube of that number and prints it.

Solution: MultiThreadMath.java

```
public class MultiThreadMath {
    static int num; static boolean ready = false;
    public static void main(String[] a) {
        Thread gen = new Thread(() -> {
            java.util.Random r = new java.util.Random();
            while(true) {
                synchronized(MultiThreadMath.class) {
                    num = r.nextInt(20)+1; ready = true;
                    System.out.println("Generated: " + num);
                    MultiThreadMath.class.notifyAll();
                    try { MultiThreadMath.class.wait(); } catch(Exception e){}
                }
                try { Thread.sleep(1000); } catch(Exception e){}
            }
        });
        Thread even = new Thread(() -> {
            while(true) { synchronized(MultiThreadMath.class) {
                try { while(!ready) MultiThreadMath.class.wait(); }
                catch(Exception e){}
                if(num%2==0) System.out.println("Square: "+(num*num));
                ready=false; MultiThreadMath.class.notifyAll();
            }
        });
        Thread odd = new Thread(() -> {
            while(true) { synchronized(MultiThreadMath.class) {
                try { while(!ready) MultiThreadMath.class.wait(); }
                catch(Exception e){}
                if(num%2!=0) System.out.println("Cube: "+(num*num*num));
                ready=false; MultiThreadMath.class.notifyAll();
            }
        });
        gen.start(); even.start(); odd.start();
    }
}
```

Q2. [15 Marks]

Write a Java program for the following:

- To create a Product (Pid, Pname, Price) table.
- Insert at least five records into the table.
- Display all the records from the table.

Solution: ProductTable.java

```
import java.sql.*;
public class ProductTable {
    public static void main(String[] a) throws Exception {
        Class.forName("com.mysql.jdbc.Driver");
        Connection con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/mydb","root","password");
        Statement st = con.createStatement();
        st.execute("CREATE TABLE IF NOT EXISTS Product("+
            "Pid INT, Pname VARCHAR(30), Price DOUBLE)");
        String[] data = {"(1, 'Pen', 10)", "(2, 'Book', 50)", "(3, 'Bag', 200)",
            "(4, 'Ruler', 15)", "(5, 'Eraser', 5)"};
        for(String d : data) st.execute("INSERT INTO Product VALUES"+d);
        ResultSet rs = st.executeQuery("SELECT * FROM Product");
        System.out.printf("%-5s %-15s %-10s\n", "Pid", "Pname", "Price");
        while(rs.next())
            System.out.printf("%-5d %-15s %-10.2f\n",
                rs.getInt(1), rs.getString(2), rs.getDouble(3));
        con.close();
    }
}
```

Q3. Viva [05 Marks]

SLIP 8

Q1. [15 Marks]

Write a Java program to define a thread for printing text on output screen for 'n' number of times. Create 3 threads and run them. Pass the text and 'n' parameters to the thread constructor.

- First thread prints "COVID19" 10 times.
- Second thread prints "LOCKDOWN2020" 20 times.
- Third thread prints "VACCINATED2021" 30 times.

Solution: PrintThread.java

```
public class PrintThread extends Thread {
    String text; int n;
    PrintThread(String t, int n) { text=t; this.n=n; }
    public void run() {
        for(int i=0;i<n;i++) System.out.println(text);
    }
    public static void main(String[] a) {
        new PrintThread("COVID19",10).start();
        new PrintThread("LOCKDOWN2020",20).start();
        new PrintThread("VACCINATED2021",30).start();
    }
}
```


Q2. [15 Marks]

Write a JSP program to check whether a given number is prime or not. Display the result in red color.

Solution: prime.jsp

```
<html><body>
<form method="post">
Enter Number: <input name="num" type="text">
<input type="submit" value="Check">
</form>
<%
String s = request.getParameter("num");
if(s != null) {
int n = Integer.parseInt(s); boolean prime = n > 1;
for(int i=2; i<=Math.sqrt(n); i++) if(n%i==0){ prime=false; break; }
out.println("<p style='color:red;'>"+n+" is "+(prime?"":"NOT ")+"Prime</p>");
}
%>
</body></html>
```

Q3. Viva [05 Marks]

SLIP 9

Q1. [15 Marks]

Write a Java program to create a thread for moving a ball inside a panel vertically. The ball should be created when the user clicks on the start button.

Solution: MovingBall.java

```
import javax.swing.*; import java.awt.*;
public class MovingBall extends JFrame {
int y = 10;
JPanel panel = new JPanel() {
protected void paintComponent(Graphics g) {
super.paintComponent(g);
g.setColor(Color.RED); g.fillOval(100,y,40,40);
}
};
public MovingBall() {
JButton btn = new JButton("Start");
btn.addActionListener(e -> new Thread(() -> {
while(true) {
y = (y+5) % (getHeight()-40);
panel.repaint();
try{Thread.sleep(30);}catch(Exception ex){}
}
}).start());
add(panel); add(btn, BorderLayout.SOUTH);
setSize(300,400); setVisible(true);
setDefaultCloseOperation(EXIT_ON_CLOSE);
}
public static void main(String[] a) { new MovingBall(); }
}
```

Q2. [15 Marks]

Write a Java program using Spring to display the message "If you can't explain it simply, you don't understand it well enough".

Solution: applicationContext.xml + Main.java

```
<!-- applicationContext.xml -->
<beans xmlns="http://www.springframework.org/schema/beans"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="...">
<bean id="msg" class="Message">
<property name="text"
value="If you can't explain it simply, you don't understand it well enough"/>
</bean>
</beans>

 
// Message.java
public class Message {
private String text;
public void setText(String t){ text=t; }
public void display(){ System.out.println(text); }
}

 
// Main.java
import org.springframework.context.support.ClassPathXmlApplicationContext;
public class Main {
public static void main(String[] a) {
new ClassPathXmlApplicationContext("applicationContext.xml")
.getBean("msg",Message.class).display();
}
}
```

Q3. Viva [05 Marks]

Q1. [15 Marks]

Write a Java program to display the Current Date using Spring.

Solution: DateBean.java + Main.java

```
// applicationContext.xml: <bean id="d" class="DateBean"/>
// DateBean.java
import java.util.Date;
public class DateBean {
    public void show() { System.out.println("Current Date: " + new Date()); }
}
// Main.java
import org.springframework.context.support.ClassPathXmlApplicationContext;
public class Main {
    public static void main(String[] a) {
        new ClassPathXmlApplicationContext("applicationContext.xml")
        .getBean("d", DateBean.class).show();
    }
}
```

Q2. [15 Marks]

Write a Java program to display first record from student table (RNo, SName, Per) onto the TextFields by clicking on button. (Assume Student table is already created.)

Solution: StudentRecord.java

```
import javax.swing.*; import java.sql.*;
public class StudentRecord extends JFrame {
    JTextField rno=new JTextField(8), sname=new JTextField(12), per=new JTextField(8);
    public StudentRecord() {
        setLayout(new java.awt.FlowLayout());
        add(new JLabel("RNo:")); add(rno);
        add(new JLabel("Name:")); add(sname);
        add(new JLabel("Per:")); add(per);
        JButton btn = new JButton("Show First Record");
        btn.addActionListener(e -> {
            try {
                Class.forName("com.mysql.jdbc.Driver");
                Connection con = DriverManager.getConnection(
                    "jdbc:mysql://localhost:3306/mydb","root","password");
                ResultSet rs = con.createStatement().executeQuery(
                    "SELECT * FROM Student LIMIT 1");
                if(rs.next()){
                    rno.setText(String.valueOf(rs.getInt(1)));
                    sname.setText(rs.getString(2));
                    per.setText(rs.getString(3));
                }
                con.close();
            } catch(Exception ex){ ex.printStackTrace(); }
        });
        add(btn); setSize(350,150); setVisible(true);
        setDefaultCloseOperation(EXIT_ON_CLOSE);
    }
    public static void main(String[] a) { new StudentRecord(); }
}
```

Q3. Viva [05 Marks]

Q1. [15 Marks]

Design an HTML page which passes customer number to a search servlet. The servlet searches for the customer number in a database (customer table) and returns customer details if found, otherwise displays an error message.

Solution: search.html + SearchServlet.java

```
<!-- search.html -->
<form action="SearchServlet" method="post">
Customer No: <input type="text" name="cno">
<input type="submit" value="Search">
</form>
&nbsp;
// SearchServlet.java
import java.io.*; import java.sql.*;
import javax.servlet.*; import javax.servlet.http.*;
public class SearchServlet extends HttpServlet {
public void doPost(HttpServletRequest req, HttpServletResponse res)
throws IOException {
res.setContentType("text/html");
PrintWriter pw = res.getWriter();
try {
Class.forName("com.mysql.jdbc.Driver");
Connection con = DriverManager.getConnection(
"jdbc:mysql://localhost:3306/mydb","root","password");
PreparedStatement ps = con.prepareStatement(
"SELECT * FROM Customer WHERE Cno=?");
ps.setInt(1, Integer.parseInt(req.getParameter("cno")));
ResultSet rs = ps.executeQuery();
if(rs.next())
pw.println("Name: "+rs.getString(2)+" | Address: "+rs.getString(3));
else pw.println("<b>Customer not found!</b>");
con.close();
} catch(Exception e){ pw.println("Error: "+e.getMessage()); }
}
}
```

Q2. [15 Marks]

Write a Java program to display information about all columns in the DONAR table using ResultSetMetaData.

Solution: MetaDataDemo.java

```
import java.sql.*;
public class MetaDataDemo {
public static void main(String[] a) throws Exception {
Class.forName("com.mysql.jdbc.Driver");
Connection con = DriverManager.getConnection(
"jdbc:mysql://localhost:3306/mydb","root","password");
ResultSet rs = con.createStatement()
.executeQuery("SELECT * FROM DONAR");
ResultSetMetaData md = rs.getMetaData();
for(int i=1; i<=md.getColumnCount(); i++)
System.out.printf("Column: %-15s | Type: %-10s | Size: %d\n",
md.getColumnName(i), md.getColumnTypeName(i),
md.getColumnDisplaySize(i));
con.close();
}
}
```

Q3. Viva [05 Marks]

Q1. [15 Marks]

Write a JSP program to check whether given number is Perfect or not. (Use Include directive).

Solution: check.jsp + perfect.jsp

```
<!-- check.jsp -->
<html><body>
<form method="post">
Enter Number: <input name="n" type="text">
<input type="submit" value="Check">
</form>
<% if(request.getParameter("n")!=null) { %>
<%@ include file="perfect.jsp" %>
<% } %>
</body></html>
&nbsp;
<!-- perfect.jsp -->
<%
int n=Integer.parseInt(request.getParameter("n")), sum=0;
for(int i=1; i<n; i++) if(n%i==0) sum+=i;
out.println("<p>"+n+" is "+(sum==n?"":"NOT ")+"Perfect</p>");
%>
```

Q2. [15 Marks]

Write a Java Program to create a PROJECT table with fields project_id, Project_name, Project_description, Project_Status. Insert values in the table. Display all the details of the PROJECT table in a tabular format on the screen (using Swing).

Solution: ProjectTable.java

```
import javax.swing.*; import java.sql.*; import java.awt.*;
public class ProjectTable extends JFrame {
public ProjectTable() throws Exception {
Class.forName("com.mysql.jdbc.Driver");
Connection con = DriverManager.getConnection(
"jdbc:mysql://localhost:3306/mydb","root","password");
Statement st = con.createStatement();
st.execute("CREATE TABLE IF NOT EXISTS PROJECT("+
"project_id INT, Project_name VARCHAR(30),"+
"Project_description VARCHAR(100), Project_Status VARCHAR(20))");
st.execute("INSERT IGNORE INTO PROJECT VALUES"+
"(1,'AI App','ML based app','Active')");
st.execute("INSERT IGNORE INTO PROJECT VALUES"+
"(2,'WebApp','Java EE app','Completed')");
ResultSet rs = st.executeQuery("SELECT * FROM PROJECT");
String[] cols = {"ID","Name","Description","Status"};
java.util.List<Object[]> rows = new java.util.ArrayList<>();
while(rs.next())
rows.add(new Object[]{rs.getInt(1),rs.getString(2),
rs.getString(3),rs.getString(4)});
JTable table = new JTable(rows.toArray(new Object[0][]), cols);
add(new JScrollPane(table));
setSize(550,200); setVisible(true);
setDefaultCloseOperation(EXIT_ON_CLOSE); con.close();
}
public static void main(String[] a) throws Exception { new ProjectTable(); }
}
```

Q3. Viva [05 Marks]

Q1. [15 Marks]

Write a Java program to display information about the database and list all the tables in the database. (Use DatabaseMetaData).

Solution: DBMetaData.java

```
import java.sql.*;
public class DBMetaData {
    public static void main(String[] a) throws Exception {
        Class.forName("com.mysql.jdbc.Driver");
        Connection con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/mydb","root","password");
        DatabaseMetaData dm = con.getMetaData();
        System.out.println("DB: "+dm.getDatabaseProductName()
            +" "+dm.getDatabaseProductVersion());
        System.out.println("Driver: "+dm.getDriverName());
        System.out.println("URL: "+dm.getURL());
        ResultSet rs = dm.getTables(null,null,"%",new String[]{"TABLE"});
        System.out.println("\nTables:");
        while(rs.next()) System.out.println(" "+rs.getString("TABLE_NAME"));
        con.close();
    }
}
```

Q2. [15 Marks]

Write a Java program to show lifecycle (creation, sleep, and dead) of a thread. Program should print randomly the name of thread and value of sleep time. The name of the thread should be hard coded through constructor. The sleep time of a thread will be a random integer in the range 0 to 4999.

Solution: ThreadLifecycle.java

```
import java.util.Random;
class MyThread extends Thread {
    MyThread(String name) { super(name); }
    public void run() {
        int sleep = new Random().nextInt(5000);
        System.out.println(getName()+" CREATED | sleep="+sleep+"ms");
        try { Thread.sleep(sleep); } catch(Exception e) {}
        System.out.println(getName()+" DEAD");
    }
}
public class ThreadLifecycle {
    public static void main(String[] a) {
        new MyThread("Alpha").start();
        new MyThread("Beta").start();
        new MyThread("Gamma").start();
    }
}
```

Q3. Viva [05 Marks]

Q1. [15 Marks]

Write a Java program for a simple search engine. Accept a string to be searched. Search the string in all text files in the current folder. Use a separate thread for each file. The result should display the filename and line number where the string is found.

Solution: SearchEngine.java

```
import java.io.*; import java.util.Scanner;
public class SearchEngine {
    static String keyword;
    static class FileSearch extends Thread {
        File file;
        FileSearch(File f) { file=f; }
        public void run() {
            try(BufferedReader br = new BufferedReader(new FileReader(file))) {
                String line; int lineNo=0;
                while((line=br.readLine())!=null) {
                    lineNo++;
                    if(line.contains(keyword))
                        System.out.println(file.getName()+":Line "+lineNo+": "+line);
                }
            } catch(Exception e){}
        }
    }
    public static void main(String[] a) throws Exception {
        Scanner sc = new Scanner(System.in);
        System.out.print("Search: "); keyword = sc.nextLine();
        for(File f : new File(".").listFiles((d,n)->n.endsWith(".txt")))
            new FileSearch(f).start();
    }
}
```

Q2. [15 Marks]

Write a JSP program to calculate sum of first and last digit of a given number. Display sum in Red Color with font size 18.

Solution: sumdigit.jsp

```
<html><body>
<form method="post">
Enter Number: <input name="num" type="text">
<input type="submit" value="Calculate">
</form>
<%
String s = request.getParameter("num");
if(s!=null && !s.isEmpty()) {
    int n = Math.abs(Integer.parseInt(s));
    int first = Integer.parseInt(String.valueOf(("" +n).charAt(0)));
    int last = n % 10;
    out.println("<p style='color:red;font-size:18px;'>"+
        "Sum of First + Last digit = "+(first+last)+"</p>");
}
%>
</body></html>
```

Q3. Viva [05 Marks]

Q1. [15 Marks]

Write a Java program to display name and priority of a Thread.

Solution: ThreadInfo.java

```
class MyThread extends Thread {
    MyThread(String name, int priority) {
        super(name); setPriority(priority);
    }
    public void run() {
        System.out.println("Name: "+getName()+" | Priority: "+getPriority());
    }
}
public class ThreadInfo {
    public static void main(String[] a) {
        new MyThread("LowThread",Thread.MIN_PRIORITY).start();
        new MyThread("NormalThread",Thread.NORM_PRIORITY).start();
        new MyThread("HighThread",Thread.MAX_PRIORITY).start();
    }
}
```

Q2. [15 Marks]

Write a SERVLET program which counts how many times a user has visited a web page. If user is visiting the page for the first time, display a welcome message. If the user is revisiting the page, display the number of times visited. (Use Cookie)

Solution: VisitCounter.java

```
import java.io.*; import javax.servlet.*; import javax.servlet.http.*;
public class VisitCounter extends HttpServlet {
    public void doGet(HttpServletRequest req, HttpServletResponse res)
        throws IOException {
        res.setContentType("text/html");
        PrintWriter pw = res.getWriter();
        Cookie[] cookies = req.getCookies(); int count = 0;
        if(cookies != null)
            for(Cookie c : cookies)
                if(c.getName().equals("visits"))
                    count = Integer.parseInt(c.getValue());
        count++;
        res.addCookie(new Cookie("visits",""+count));
        if(count == 1) pw.println("<h2>Welcome! First visit!</h2>");
        else pw.println("<h2>You have visited "+count+" times.</h2>");
    }
}
```

Q3. Viva [05 Marks]

Q1. [15 Marks]

Write a Java program to create a TreeSet, add some colors (String) and print out the content of TreeSet in ascending order.

Solution: ColorSet.java

```
import java.util.TreeSet;
public class ColorSet {
    public static void main(String[] a) {
        TreeSet<String> ts = new TreeSet<>();
        ts.add("Red"); ts.add("Blue"); ts.add("Green");
        ts.add("Pink"); ts.add("Yellow"); ts.add("Orange");
        System.out.println("Colors in ascending order: " + ts);
    }
}
```

Q2. [15 Marks]

Write a Java program to accept the details of Teacher (TNo, TName, Subject). Insert at least 5 Records into Teacher Table and display the details of Teacher who is teaching "JAVA" Subject. (Use PreparedStatement Interface)

Solution: TeacherDB.java

```
import java.sql.*;
public class TeacherDB {
    public static void main(String[] a) throws Exception {
        Class.forName("com.mysql.jdbc.Driver");
        Connection con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/mydb","root","password");
        Statement st = con.createStatement();
        st.execute("CREATE TABLE IF NOT EXISTS Teacher(" +
            "TNo INT, TName VARCHAR(30), Subject VARCHAR(30))");
        String[][] data = {{ "1", "Alice", "JAVA"}, {"2", "Bob", "C++"},
            {"3", "Charlie", "JAVA"}, {"4", "Dave", "Python"}, {"5", "Eve", "JAVA"} };
        PreparedStatement ps = con.prepareStatement(
            "INSERT INTO Teacher VALUES(?,?,?)");
        for(String[] d : data){
            ps.setInt(1,Integer.parseInt(d[0]));
            ps.setString(2,d[1]); ps.setString(3,d[2]);
            ps.executeUpdate();
        }
        ps = con.prepareStatement(
            "SELECT * FROM Teacher WHERE Subject=?");
        ps.setString(1,"JAVA");
        ResultSet rs = ps.executeQuery();
        System.out.println("Teachers teaching JAVA:");
        while(rs.next())
            System.out.println(rs.getInt(1)+" | "+rs.getString(2));
        con.close();
    }
}
```

Q3. Viva [05 Marks]

Q1. [15 Marks]

Write a Java program to accept 'N' integers from a user. Store and display integers in sorted order having proper collection class. The collection should not accept duplicate elements.

Solution: SortedIntegers.java

```
import java.util.*; import java.util.Scanner;
public class SortedIntegers {
    public static void main(String[] a) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter N: "); int n = sc.nextInt();
        TreeSet<Integer> ts = new TreeSet<>();
        for(int i=0;i<n;i++){
            System.out.print("Enter integer: "); ts.add(sc.nextInt());
        }
        System.out.println("Sorted unique integers: " + ts);
    }
}
```

Q2. [15 Marks]

Write a Multithreading program in Java to display the numbers between 1 to 100 continuously in a TextField by clicking on button. (Use Runnable Interface).

Solution: CounterApp.java

```
import javax.swing.*; import java.awt.*;
public class CounterApp extends JFrame implements Runnable {
    JTextField tf = new JTextField(10);
    public CounterApp() {
        JButton btn = new JButton("Start");
        btn.addActionListener(e -> new Thread(this).start());
        add(new JLabel("Number:"), BorderLayout.WEST);
        add(tf, BorderLayout.CENTER);
        add(btn, BorderLayout.EAST);
        setSize(300,80); setVisible(true);
        setDefaultCloseOperation(EXIT_ON_CLOSE);
    }
    public void run() {
        for(int i=1; i<=100; i++) {
            final int x=i;
            SwingUtilities.invokeLater(() -> tf.setText(""+x));
            try{ Thread.sleep(100); }catch(Exception e){}
        }
    }
    public static void main(String[] a) { new CounterApp(); }
}
```

Q3. Viva [05 Marks]

SLIP 18

Q1. [15 Marks]

Write a Java program to display name and priority of a Thread.

Solution: ThreadInfo.java

```
// (Same structure as Slip 15 Q1)
class MyThread extends Thread {
    MyThread(String name, int p) { super(name); setPriority(p); }
    public void run() {
        System.out.println("Name: "+getName()+" | Priority: "+getPriority());
    }
}

public class ThreadInfo {
    public static void main(String[] a) {
        new MyThread("T1",Thread.MIN_PRIORITY).start();
        new MyThread("T2",Thread.NORM_PRIORITY).start();
        new MyThread("T3",Thread.MAX_PRIORITY).start();
    }
}
```

Q2. [15 Marks]

Write a SERVLET program in Java to accept details of student (SeatNo, Stud_Name, Class, Total_Marks). Calculate percentage and grade obtained and display details on page.

Solution: StudentServlet.java

```
import java.io.*; import javax.servlet.*; import javax.servlet.http.*;
public class StudentServlet extends HttpServlet {
    public void doPost(HttpServletRequest req, HttpServletResponse res)
        throws IOException {
        res.setContentType("text/html");
        PrintWriter pw = res.getWriter();
        String sno=req.getParameter("seatNo");
        String name=req.getParameter("name");
        String cls=req.getParameter("cls");
        double total=Double.parseDouble(req.getParameter("total"));
        double per = total / 5.0; // assuming 5 subjects, max 100 each
        String grade = per>=75?"A":per>=60?"B":per>=50?"C":per>=35?"D":"F";
        pw.println("<h2>Student Result</h2>");
        pw.println("Seat No: "+sno+"<br>Name: "+name+"<br>Class: "+cls);
        pw.println("<br>Total: "+total+"<br>Percentage: "+per+"%");
        pw.println("<br><b>Grade: "+grade+"</b>");
    }
}
```

Q3. Viva [05 Marks]

Q1. [15 Marks]

Write a Java program to accept 'N' Integers from a user store them into LinkedList Collection and display only negative integers.

Solution: NegativeIntegers.java

```
import java.util.*; import java.util.Scanner;
public class NegativeIntegers {
    public static void main(String[] a) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter N: "); int n = sc.nextInt();
        LinkedList<Integer> list = new LinkedList<>();
        for(int i=0;i<n;i++){
            System.out.print("Enter: "); list.add(sc.nextInt());
        }
        System.out.print("Negative integers: ");
        list.stream().filter(x -> x < 0).forEach(x -> System.out.print(x+" "));
    }
}
```

Q2. [15 Marks]

Write a SERVLET application to accept username and password, search them into database, if found then display appropriate message on the browser otherwise display error message.

Solution: LoginServlet.java

```
import java.io.*; import java.sql.*;
import javax.servlet.*; import javax.servlet.http.*;
public class LoginServlet extends HttpServlet {
    public void doPost(HttpServletRequest req, HttpServletResponse res)
        throws IOException {
        res.setContentType("text/html");
        PrintWriter pw = res.getWriter();
        try {
            Class.forName("com.mysql.jdbc.Driver");
            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/mydb","root","password");
            PreparedStatement ps = con.prepareStatement(
                "SELECT * FROM Users WHERE username=? AND password=?");
            ps.setString(1, req.getParameter("username"));
            ps.setString(2, req.getParameter("password"));
            if(ps.executeQuery().next())
                pw.println("<h2 style='color:green;'>Login Successful!</h2>");
            else
                pw.println("<h2 style='color:red;'>Invalid credentials!</h2>");
            con.close();
        } catch(Exception e){ pw.println("Error: "+e.getMessage()); }
    }
}
```

Q3. Viva [05 Marks]

Q1. [15 Marks]

Create a JSP page to accept a number from a user and display it in words. Example: 123 – One Two Three. The output should be in red color.

Solution: numwords.jsp

```
<html><body>
<form method="post">
Enter Number: <input name="num" type="text">
<input type="submit" value="Convert">
</form>
<%
String s = request.getParameter("num");
if(s != null) {
String[] words = {"Zero", "One", "Two", "Three", "Four",
"Five", "Six", "Seven", "Eight", "Nine"};
StringBuilder sb = new StringBuilder();
for(char c : s.toCharArray())
if(Character.isDigit(c)) sb.append(words[c-'0']+" ");
out.println("<p style='color:red;'>" + sb.toString().trim() + "</p>");
}
%>
</body></html>
```

Q2. [15 Marks]

Write a Java program to blink image on the JFrame continuously.

Solution: BlinkImage.java

```
import javax.swing.*; import java.awt.*;
public class BlinkImage extends JFrame {
ImageIcon img = new ImageIcon("image.png"); // place image.png in project
JLabel lbl = new JLabel(img);
public BlinkImage() {
add(lbl); setSize(300,300); setVisible(true);
setDefaultCloseOperation(EXIT_ON_CLOSE);
new Thread(() -> {
while(true) {
lbl.setVisible(!lbl.isVisible());
try{ Thread.sleep(500); }catch(Exception e){}
}
}).start();
}
public static void main(String[] a) { new BlinkImage(); }
}
```

Q3. Viva [05 Marks]

SLIP 21

Q1. [15 Marks]

Write a Java program to accept 'N' Subject Names from a user store them into LinkedList Collection and Display them by using Iterator interface.

Solution: SubjectList.java

```
import java.util.*; import java.util.Scanner;
public class SubjectList {
public static void main(String[] a) {
Scanner sc = new Scanner(System.in);
System.out.print("Enter N: "); int n = sc.nextInt(); sc.nextLine();
LinkedList<String> list = new LinkedList<>();
for(int i=0;i<n;i++){
System.out.print("Subject: "); list.add(sc.nextLine());
}
System.out.println("Subjects (using Iterator):");
Iterator<String> it = list.iterator();
while(it.hasNext()) System.out.println(it.next());
}
}
```

Q2. [15 Marks]

Write a Java program to solve producer consumer problem in which a producer produces a value and consumer consumes the value before producer generates the next value. (Hint: use thread synchronization)

Solution: ProducerConsumer.java

```
class SharedBuffer {
    int data; boolean hasData = false;
    synchronized void produce(int val) throws InterruptedException {
        while(hasData) wait();
        data = val; hasData = true;
        System.out.println("Produced: " + val);
        notifyAll();
    }
    synchronized void consume() throws InterruptedException {
        while(!hasData) wait();
        System.out.println("Consumed: " + data);
        hasData = false; notifyAll();
    }
}

public class ProducerConsumer {
    public static void main(String[] a) {
        SharedBuffer buf = new SharedBuffer();
        new Thread(() -> {
            try{ for(int i=1;i<=5;i++) buf.produce(i); }
            catch(Exception e){}
        }).start();
        new Thread(() -> {
            try{ for(int i=1;i<=5;i++) buf.consume(); }
            catch(Exception e){}
        }).start();
    }
}
```

Q3. Viva [05 Marks]

Q1. [15 Marks]

Write a Menu Driven program in Java for the following: Assume Employee table with attributes (ENo, EName, Salary) is already created.

1. Insert 2. Update 3. Display 4. Exit

Solution: EmployeeMenu.java

```
import java.sql.*; import java.util.Scanner;
public class EmployeeMenu {
    static Connection con;
    public static void main(String[] a) throws Exception {
        Class.forName("com.mysql.jdbc.Driver");
        con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/mydb","root","password");
        Scanner sc = new Scanner(System.in); int ch;
        do {
            System.out.println("1.Insert 2.Update 3.Display 4.Exit");
            ch = sc.nextInt();
            if(ch==1) {
                System.out.print("ENo EName Salary: ");
                con.createStatement().execute(
                    "INSERT INTO Employee VALUES("+sc.nextInt()+", '"+sc.next()+"', "+sc.nextDouble()+")");
                System.out.println("Record inserted.");
            } else if(ch==2) {
                System.out.print("New Salary and ENo: ");
                double sal=sc.nextDouble(); int eno=sc.nextInt();
                con.createStatement().execute(
                    "UPDATE Employee SET Salary="+sal+" WHERE ENo="+eno);
                System.out.println("Updated.");
            } else if(ch==3) {
                ResultSet rs = con.createStatement()
                    .executeQuery("SELECT * FROM Employee");
                while(rs.next())
                    System.out.println(rs.getInt(1)+" "+rs.getString(2)+" "+rs.getDouble(3));
            }
        } while(ch != 4);
        con.close();
    }
}
```

Q2. [15 Marks]

Write a JSP program which accepts UserName in a TextBox and greets the user according to the time on server machine.

Solution: greet.jsp

```
<html><body>
<form method="post">
    Username: <input name="uname" type="text">
    <input type="submit" value="Greet">
</form>
<%
    String uname = request.getParameter("uname");
    if(uname != null && !uname.isEmpty()) {
        int hour = new java.util.Date().getHours();
        String greet = hour<12 ? "Good Morning"
            : hour<17 ? "Good Afternoon"
            : "Good Evening";
        out.println("<h2>"+greet+" "+uname+"!</h2>");
    }
%>
</body></html>
```

Q3. Viva [05 Marks]

Q1. [15 Marks]

Write a Java program to accept a String from a user and display each vowel from a String after every 3 seconds.

Solution: VowelDisplay.java

```
import java.util.Scanner;
public class VowelDisplay extends Thread {
    String str;
    VowelDisplay(String s) { str = s.toLowerCase(); }
    public void run() {
        for(char c : str.toCharArray()) {
            if("aeiou".indexOf(c) >= 0) {
                System.out.println("Vowel: " + c);
                try{ Thread.sleep(3000); }catch(Exception e){}
            }
        }
    }
    public static void main(String[] a) {
        System.out.print("Enter string: ");
        String s = new Scanner(System.in).nextLine();
        new VowelDisplay(s).start();
    }
}
```

Q2. [15 Marks]

Write a Java program to accept 'N' student names through command line, store them into the appropriate Collection and display them by using Iterator and ListIterator interface.

Solution: StudentNames.java

```
import java.util.*;
public class StudentNames {
    public static void main(String[] args) {
        // Pass names via command line: java StudentNames Alice Bob Charlie
        LinkedList<String> list = new LinkedList<>(Arrays.asList(args));
        System.out.println("Forward (Iterator):");
        Iterator<String> it = list.iterator();
        while(it.hasNext()) System.out.println(it.next());
        System.out.println("\nBackward (ListIterator):");
        ListIterator<String> li = list.listIterator(list.size());
        while(li.hasPrevious()) System.out.println(li.previous());
    }
}
```

Q3. Viva [05 Marks]

Q1. [15 Marks]

Write a Java program to scroll the text from left to right continuously.

Solution: ScrollText.java

```
import javax.swing.*; import java.awt.*;
public class ScrollText extends JFrame {
    int x = -200;
    public ScrollText() {
        setSize(500,100); setVisible(true);
        setDefaultCloseOperation(EXIT_ON_CLOSE);
        new Thread(() -> {
            while(true) {
                x = (x+3) % getWidth();
                repaint();
                try{ Thread.sleep(30); }catch(Exception e){}
            }
        }).start();
    }
    public void paint(Graphics g) {
        super.paint(g);
        g.setFont(new Font("Arial", Font.BOLD, 20));
        g.drawString("Scrolling Text...", x, 60);
    }
    public static void main(String[] a) { new ScrollText(); }
}
```

Q2. [15 Marks]

Write a JSP script to accept username and password from user, if they are same then display "Login Successfully" message in Login.html file, otherwise display "Login Failed" Message in Error.html file.

Solution: login.jsp

```
<!-- login.jsp -->
<html><body>
<form method="post">
Username: <input name="username" type="text"><br>
Password: <input name="password" type="password"><br>
<input type="submit" value="Login">
</form>
<%
String user = request.getParameter("username");
String pass = request.getParameter("password");
if(user != null) {
if(user.equals(pass)) response.sendRedirect("Login.html");
else response.sendRedirect("Error.html");
}
%>
</body></html>
```

Q3. Viva [05 Marks]

Q1. [15 Marks]

Write a JSP program to accept Name and Age of Voter and check whether he is eligible for voting or not.

Solution: voter.jsp

```
<html><body>
<form method="post">
Name: <input name="name" type="text"><br>
Age: <input name="age" type="number"><br>
<input type="submit" value="Check">
</form>
<%
String name = request.getParameter("name");
String age = request.getParameter("age");
if(age != null) {
int a = Integer.parseInt(age);
out.println("<p>" + name + " is " + (a>=18?"":"NOT ") +
"eligible to vote.</p>");
}
%>
</body></html>
```

Q2. [15 Marks]

Write a Java Program (Swing) to execute DDL queries (Create Table, Alter Table, Drop Table) entered by the user in a text field.

Solution: DDLExecutor.java

```
import javax.swing.*; import java.sql.*;
public class DDLExecutor extends JFrame {
JTextField tf = new JTextField(22);
public DDLExecutor() {
setLayout(new java.awt.FlowLayout());
add(new JLabel("DDL Query:")); add(tf);
JButton create=new JButton("Create Table");
JButton alter =new JButton("Alter Table");
JButton drop =new JButton("Drop Table");
create.addActionListener(e -> exec(tf.getText()));
alter .addActionListener(e -> exec(tf.getText()));
drop .addActionListener(e -> exec(tf.getText()));
add(create); add(alter); add(drop);
setSize(420,120); setVisible(true);
setDefaultCloseOperation(EXIT_ON_CLOSE);
}
void exec(String q) {
try {
Class.forName("com.mysql.jdbc.Driver");
Connection con = DriverManager.getConnection(
"jdbc:mysql://localhost:3306/mydb","root","password");
con.createStatement().execute(q);
JOptionPane.showMessageDialog(this,"Executed Successfully!");
con.close();
} catch(Exception e){
JOptionPane.showMessageDialog(this,"Error: "+e.getMessage());
}
}
public static void main(String[] a) { new DDLExecutor(); }
}
```

Q3. Viva [05 Marks]

Q1. [15 Marks]

Write a Java program to delete the details of given employee (ENo, EName, Salary). Accept employee ID through command line. (Use PreparedStatement Interface)

Solution: DeleteEmployee.java

```
import java.sql.*;
public class DeleteEmployee {
    public static void main(String[] a) throws Exception {
        if(a.length == 0){
            System.out.println("Usage: java DeleteEmployee <ENo>"); return;
        }
        int eno = Integer.parseInt(a[0]);
        Class.forName("com.mysql.jdbc.Driver");
        Connection con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/mydb","root","password");
        PreparedStatement ps = con.prepareStatement(
            "DELETE FROM Employee WHERE ENo=?");
        ps.setInt(1, eno);
        int rows = ps.executeUpdate();
        System.out.println(rows>0 ? "Employee "+eno+" deleted."
            : "Employee not found.");
        con.close();
    }
}
```

Q2. [15 Marks]

Write a JSP program to calculate sum of first and last digit of a given number. Display sum in Red Color with font size 18.

Solution: sumdigit.jsp

```
<html><body>
<form method="post">
Enter Number: <input name="num" type="text">
<input type="submit" value="Calculate">
</form>
<%
String s = request.getParameter("num");
if(s != null && !s.isEmpty()) {
int n = Math.abs(Integer.parseInt(s));
int first = Integer.parseInt(String.valueOf(""+n).charAt(0));
int last = n % 10;
out.println("<p style='color:red;font-size:18px;'>"+
"Sum = "+(first+last)+"</p>");
}
%>
</body></html>
```

Q3. Viva [05 Marks]

Q1. [15 Marks]

Write a Java Program to display the details of College (CID, CName, Address, Year) on JTable.

Solution: CollegeTable.java

```
import javax.swing.*; import java.sql.*;
public class CollegeTable extends JFrame {
    public CollegeTable() throws Exception {
        Class.forName("com.mysql.jdbc.Driver");
        Connection con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/mydb","root","password");
        ResultSet rs = con.createStatement()
            .executeQuery("SELECT * FROM College");
        String[] cols = {"CID", "CName", "Address", "Year"};
        java.util.List<Object[]> rows = new java.util.ArrayList<>();
        while(rs.next())
            rows.add(new Object[]{rs.getInt(1),rs.getString(2),
                rs.getString(3),rs.getInt(4)});
        JTable table = new JTable(rows.toArray(new Object[0][]), cols);
        add(new JScrollPane(table));
        setSize(500,300); setVisible(true);
        setDefaultCloseOperation(EXIT_ON_CLOSE); con.close();
    }
    public static void main(String[] a) throws Exception { new CollegeTable(); }
}
```

Q2. [15 Marks]

Write a SERVLET program to change inactive time interval of session.

Solution: SessionTimeout.java

```
import java.io.*; import javax.servlet.*; import javax.servlet.http.*;
public class SessionTimeout extends HttpServlet {
    public void doGet(HttpServletRequest req, HttpServletResponse res)
        throws IOException {
        res.setContentType("text/html");
        HttpSession session = req.getSession();
        session.setMaxInactiveInterval(60); // 60 seconds
        res.getWriter().println(
            "<h2>Session timeout set to 60 seconds.</h2>" +
            "Session ID: " + session.getId());
    }
}
```

Q3. Viva [05 Marks]

Q1. [15 Marks]

Write a JSP script to accept a String from a user and display it in reverse order.

Solution: reverse.jsp

```
<html><body>
<form method="post">
Enter String: <input name="str" type="text">
<input type="submit" value="Reverse">
</form>
<%
String s = request.getParameter("str");
if(s != null)
out.println("<p>Reversed: " + new StringBuilder(s).reverse() + "</p>");
%>
</body></html>
```

Q2. [15 Marks]

Write a Java program to display name of currently executing Thread in multithreading.

Solution: CurrentThread.java

```
public class CurrentThread extends Thread {
    CurrentThread(String name) { super(name); }
    public void run() {
        System.out.println("Currently Running: "
            + Thread.currentThread().getName());
    }
    public static void main(String[] a) {
        new CurrentThread("Thread-A").start();
        new CurrentThread("Thread-B").start();
        new CurrentThread("Thread-C").start();
    }
}
```

Q3. Viva [05 Marks]**Q1. [15 Marks]**

Write a Java program to display information about all columns in the DONAR table using ResultSetMetaData.

Solution: MetaDataDemo.java

```
import java.sql.*;
public class MetaDataDemo {
    public static void main(String[] a) throws Exception {
        Class.forName("com.mysql.jdbc.Driver");
        Connection con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/mydb","root","password");
        ResultSet rs = con.createStatement()
            .executeQuery("SELECT * FROM DONAR");
        ResultSetMetaData md = rs.getMetaData();
        int count = md.getColumnCount();
        System.out.printf("%-5s %-20s %-15s %-8s\n",
            "No", "Column Name", "Data Type", "Size");
        for(int i=1; i<=count; i++)
            System.out.printf("%-5d %-20s %-15s %-8d\n", i,
                md.getColumnName(i), md.getColumnTypeName(i),
                md.getColumnDisplaySize(i));
        con.close();
    }
}
```

Q2. [15 Marks]

Write a Java program to create LinkedList of integer objects and perform the following:

- i. Add element at first position
- ii. Delete last element
- iii. Display the size of link list

Solution: LinkedListOps.java

```
import java.util.LinkedList;
public class LinkedListOps {
    public static void main(String[] a) {
        LinkedList<Integer> list = new LinkedList<>();
        list.addFirst(10); list.addFirst(20); list.addFirst(30);
        System.out.println("After addFirst(10,20,30): " + list);
        list.removeLast(); // ii. Delete last
        System.out.println("After removeLast: " + list);
        System.out.println("Size: " + list.size()); // iii. Size
    }
}
```

Q3. Viva [05 Marks]

SLIP 30

Q1. [15 Marks]

Write a Java program for the implementation of synchronization.

Solution: SyncDemo.java

```
class Printer {
    synchronized void print(String msg) {
        System.out.print("[");
        try{ Thread.sleep(100); }catch(Exception e){}
        System.out.println(msg + "]");
    }
}

public class SyncDemo {
    public static void main(String[] a) {
        Printer p = new Printer();
        // Without sync, output would be interleaved
        new Thread(() -> p.print("Thread 1")).start();
        new Thread(() -> p.print("Thread 2")).start();
        new Thread(() -> p.print("Thread 3")).start();
    }
}
```

Q2. [15 Marks]

Write a Java Program for the implementation of scrollable ResultSet. Assume Teacher table with attributes (TID, TName, Salary) is already created.

Solution: ScrollableRS.java

```
import java.sql.*;
public class ScrollableRS {
    public static void main(String[] a) throws Exception {
        Class.forName("com.mysql.jdbc.Driver");
        Connection con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/mydb","root","password");
        Statement st = con.createStatement(
            ResultSet.TYPE_SCROLL_INSENSITIVE,
            ResultSet.CONCUR_READ_ONLY);
        ResultSet rs = st.executeQuery("SELECT * FROM Teacher");
        System.out.println("--- Forward Traversal ---");
        while(rs.next())
            System.out.println(rs.getInt(1)+" "+rs.getString(2)+" "+rs.getDouble(3));
        System.out.println("\n--- Last Record ---");
        rs.last();
        System.out.println(rs.getInt(1)+" "+rs.getString(2)+" "+rs.getDouble(3));
        System.out.println("\n--- First Record ---");
        rs.first();
        System.out.println(rs.getInt(1)+" "+rs.getString(2)+" "+rs.getDouble(3));
        System.out.println("\n--- Backward Traversal ---");
        rs.afterLast();
        while(rs.previous())
            System.out.println(rs.getInt(1)+" "+rs.getString(2)+" "+rs.getDouble(3));
        con.close();
    }
}
```

Q3. Viva [05 Marks]
